OBSERVATION CHECKLIST

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

Unit: Computer Repair and Maintenance | **Unit Code:** 0619 451 06A

Element 3: Test Computer Components | **Module 2, Term 3**

Time: 2 hours

Candidate name	
Admission number	
Date	
Assessor	

CUTTING LIST OF MATERIALS

No.	Item	Quantity per candidate	Provided by

1	Computer with a hard disk drive, running Windows	1	Institution
2	Internet access (to download the software)	1 connection	Institution
3	CrystalDiskInfo software (installer .exe or portable ZIP edition)	1 copy	Candidate (downloaded from the official website)
4	Flash disk or folder for saving screenshots	1	Candidate
5	Results Recording Sheet	1	Institution
6	Pen	1	Candidate
7	Protective gear (overall, gloves)	1 set	Candidate
8	Task paper (Element 3: Test Computer Components)	1	Institution

Instructions to the assessor: Observe the candidate while they work. Award the allocated mark (or part of it where the item has more than one mark) when the item is met, and 0 where it is not met. Enter the score and any remarks in the columns provided.

PART 1: PRACTICAL CHECKLIST (30 MARKS)

Task 1: Preparation and Safety

No.	Item to be observed	Allocated marks	Candidate score	Remarks
1.1	Puts on protective gear, observes ESD precautions, prepares the work area, then switches on the computer and logs in	1		
1.2	Identifies the storage type (HDD or SSD) and states how it was identified	1		
1.3	Closes unnecessary programs, opens Disk Management and notes the number of drives and partitions	1		
1.4	States two correct precautions to observe before testing a storage device	2		
	Sub-total	**5**		

Task 2: Download and Install the Software

No.	Item to be observed	Allocated marks	Candidate score	Remarks
2.1	Opens a browser and goes to the official CrystalDiskInfo website	1		
2.2	Downloads the software and notes the file name and version	1		
2.3	Installs (or extracts the ZIP) and opens the program; takes a screenshot	1		

2.4	Names the website used and explains why an official source is important	1		
	Sub-total	**4**		

Task 3: Identify the Hard Disk Drive

No.	Item to be observed	Allocated marks	Candidate score	Remarks
3.1	Records the drive model name, capacity and firmware version correctly	2		
3.2	Records the interface type, rotation rate (RPM) and drive letters correctly	1		
3.3	Takes a screenshot of the main CrystalDiskInfo window	1		
	Sub-total	**4**		

Task 4: Test the Health Status of the Drive

No.	Item to be observed	Allocated marks	Candidate score	Remarks

4.1	Records the health status and percentage (if shown)	1		
4.2	Correctly states the meaning of the health indicator colours (Blue, Yellow, Red, Grey)	1		
4.3	Records the current temperature, Power On Hours and Power On Count	1		
4.4	Explains correctly what these values show about how much the drive has been used	2		
	Sub-total	**5**		

Task 5: Analyse the S.M.A.R.T. Attributes

No.	Item to be observed	Allocated marks	Candidate score	Remarks
5.1	Records the Current, Worst, Threshold and Raw values of all five attributes (Reallocated Sectors, Current Pending Sector, Uncorrectable Sector, Spin Retry, Temperature)	2		
5.2	Takes a screenshot of the S.M.A.R.T. attributes list	1		
5.3	Correctly states what a rising Reallocated Sectors Count indicates	1		
5.4	Correctly states the action to take if a Current value falls below its Threshold	1		

	Sub-total	**5**		

Task 6: Interpret Results and Make Recommendations

No.	Item to be observed	Allocated marks	Candidate score	Remarks
6.1	States the condition of the drive (good, needs monitoring or should be replaced) and gives two valid reasons that match the results	2		
6.2	States three correct maintenance practices that extend the life of an HDD	1		
6.3	States two correct signs, other than software results, that show an HDD is failing	1		
6.4	Explains why data must be backed up before any hardware repair or disk test	1		
	Sub-total	**5**		

Task 7: Document the Results and Close Down

No.	Item to be observed	Allocated marks	Candidate score	Remarks
7.1	Saves the results using the Copy option or saved screenshots	1		

7.2	Closes the program without changing any drive settings or deleting files, leaves the work area clean and tidy, and informs the assessor on finishing	1		
	Sub-total	**2**		

Practical Summary

Task	Allocated marks	Candidate score
Task 1	5	
Task 2	4	
Task 3	4	
Task 4	5	
Task 5	5	
Task 6	5	
Task 7	2	
TOTAL PRACTICAL	**30**	

PART 2: PRODUCT CHECKLIST (5 MARKS)

No.	Item to be observed	Allocated marks	Candidate score	Remarks

P1	Results Recording Sheet is fully completed (Sections A, B, C and D) with no blank entries, is neat and legible, and has the candidate's name, signature and date	2		
P2	Recorded values match what is shown in the screenshots	1		
P3	All required screenshots are submitted (program opened, main window, S.M.A.R.T. attributes)	1		
P4	Conclusion has one box ticked, with reasons and recommendations that follow from the results	1		
	TOTAL PRODUCT	**5**		

PART 3: ORAL CHECKLIST

Ask each question without hints. Award the marks shown for each correct point.

No.	Question	Expected Answer	Marks	Score	

1	What is POST and what runs it?	A self-test that checks hardware at startup (1). Run by the BIOS/UEFI (1).	2		
2	Name two parts checked during POST.	Any two: CPU, RAM, video card, keyboard, drives, motherboard (½ each).	1		
3	What does the CMOS battery do? State one sign that it has failed.	Keeps BIOS settings and the clock (1). Wrong date/time or "CMOS checksum error" (1).	2		
4	Name two ways POST reports faults.	Any two: beep codes, error messages, LEDs, display codes (1 each).	2		
5	What does each mean? (a) No beep, no display, no fan. (b) Fans spin but no display. (c) "No boot device found."	(a) Power problem (1). (b) CPU, RAM, motherboard or video fault (1). (c) Drive problem or wrong boot order (1).	3		
6	State four reasons why component testing is important.	Any four: finds the faulty part, saves money, confirms the repair, protects data, prevents repeat faults, safety (½ each).	2		
7	State two times when testing should be done.	Any two: before, during or after repair, before delivery (½ each).	1		

8	What must you do to the computer before a visual inspection? Name two things to look for.	Switch off and unplug (1). Any two: loose cables, dust, burn marks, swollen capacitors (½ each).	2		
9	What is swap testing? What does it mean if the fault disappears?	Replacing a suspected part with a known-good part (1). The old part is faulty (1).	2		
10	What is minimal configuration testing?	Starting the PC with only the basic parts (1), then adding parts back one at a time to find the faulty one (1).	2		
11	What is each tool used for? (a) Multimeter. (b) PSU tester. (c) POST card.	(a) Measures voltage and continuity (1). (b) Tests power supply outputs (1). (c) Shows POST codes when there is no display (1).	3		
12	In CrystalDiskInfo, what do blue, yellow and red mean? Name two attributes that should be 0.	Blue = Good, yellow = Caution, red = Bad (1). Any two: Reallocated Sectors, Pending Sectors, Uncorrectable Sectors (1). Drive temperature should be below 50°C (1).	3		
13	How do you test RAM with MemTest86?	Boot from a MemTest86 USB (1). Run at least 4 passes; any error means faulty RAM (1).	2		

14	What are healthy CPU temperatures at idle and under load? When do you stop Prime95?	Idle 30–50°C (½). Load below 80–85°C (½). Stop above about 95°C or if the PC freezes or restarts (1).	2		
15	In what order do you run CrystalDiskInfo, MemTest86 and Prime95?	CrystalDiskInfo, then MemTest86, then Prime95 last (1).	1		
	TOTAL		**30**		

OVERALL SUMMARY

Component	Allocated marks	Candidate score
Practical checklist	30	
Product checklist	5	
Oral checklist		
GRAND TOTAL		

DECISION OUTCOME

Decision (tick one)

☐ Competent ☐ Not yet competent

Assessor's comments / feedback:

……

……

……

CANDIDATE'S COMMENTS

(To be completed by the candidate after the assessment.)

I was informed of the assessment decision: ☐ Yes ☐ No

I am satisfied with the assessment process and decision: ☐ Yes ☐ No

Candidate's comments:

……

……

……

……

……

……

SIGNATURES

	Name	Signature	Date
Assessor			
Candidate			
Internal verifier (if applicable)			